Estimating in Building Construction

Chapter 8 Equipment

Equipment Costs

- Operating costs
  - Fuel
  - Lubrication
  - Tires
- Fixed (ownership) costs
  - Interest
  - Depreciation
  - Repairs
  - Insurance
  - Taxes
  - Storage

Fuel Cost

Fuel cost

= hp rating × Power utilization × Use factor × Consumption rate × Fuel cost

- Consumption rate:
  - Diesel 0.04-0.06 gallons per hp per hour

Example 8.1 Estimate Fuel Cost

A loader of 120-horsepower

Equipment operation (use factor) = 45 minute per hour = 0.75

Operating at (power utilization) = 70% of it rated horsepower

Fuel cost = $1.1 per gallon

Consumption rate = 0.06 gallons per hp per hour

Fuel cost = hp rating × Power utilization × Use factor
× Consumption rate × Fuel cost

Fuel cost per machine hour = 120 x 0.70 x 0.75 x 0.06 x 1.1
= $4.158

Lubrication

- Oil
- Labor

Example 8.2 Equipment Lubrication

Oil change and greased every 120 hours

Requires 6 quarts of oil for the change

Time required for oil change and greasing 2.5 hours

Oil cost = $ 1.3 per quart

Oiler labor rate = $17.5 per hour

Lubrication cost = 6 qts. Of oil x $1.3 per qt. = $7.80

Labor cost = 2.5 hours x $17.5 per labor hour = $43.75

Total cost of oil change = $7.80 + $43.75 = $51.55

Cost per machine work hour = $51.55/120 hours

= $0.43 per hour

Tires

- Depreciation
  = Cost / Useful life

- Repair
  - Typically 15% of depreciation

Example 8.3 Tires

Four tires for a piece of equipment cost $5,000

Useful life = 3,500 hours

Cost for repair to the tires = 15% of depreciation

Average cost of tires per hour = ?

Tires depreciation = $5,000/3,500 hours = $1.43 per hour

Tire repair = 15/100 x $1.43 per hour = $0.21 per hour

Tire cost = $1.43 + $0.21

= $1.64 per machine work hour

Depreciation

- Depreciable cost
  = Original cost – Salvage value

- Depreciable cost per year
  = Depreciable cost / Useful life

Example 8.4 Depreciation

Purchase price = $67,500

Anticipated salvage value = $10,000

Estimated life = 5 years

If the equipment last for = 10,000 machine work hours

Depreciable cost = Original cost – Salvage value

= 67,500 – 10,000 = 57,500

Depreciable cost per year = Depreciable cost / Useful life

= 57,500 / 5

= 11,500 per year

Depreciable cost per hour = 57,500 / 10,000

= $5.7 per machine work hour

Interest

Approximate interest cost

$$= C \times I \times L/(2H)$$

- C = Amount of loan
- I = Interest rate
- L = Life of loan
- H = Useful life of equipment (hours)

Ownership Costs

- Depreciation
- Repair, overhaul
- Interest
- Insurance
- Taxes and storage

Repairs

- Percentage of equipment cost spread over useful life

Insurance, Taxes, and Storage

- Percentage of equipment cost spread over useful life

Example 8.5 Cost of Ownership

Assumptions

Actual cost = $47,600

Horsepower rating = 150 hp

Cost of tires = $4,500

Salvage value = 3 %

Useful life = 7 years or 14,000 hours

Total interest = 8% per year

Length of loan = 7 years

Total insurance, taxes, and storage = 6% per year

Fuel cost = $1.1 per gallon

Consumption rate = 0.06 gallons per hp per hour

Power utilization = 62%

Use factor = 70 %

Cost of Ownership

Assumptions

Lubrication (Oil change) = 4 quarts oil at $1.25

Oiler labor = 2 hours at $16.50

Lubrication schedule = every 150 hours

Life of tires = 4,000 hours

Repair to tires = 12% of depreciation

Repairs to equipment = 65% over useful life

Estimate the cost of owning and operating = ?

Approximate interest cost = (47,600 x 0.08 x 7) / 2 x 14,000

= 0.95 per equip work hour

Salvage value = 47,600 x 0.03 = $1,428

Depreciation cost per hour = (47,600 – 1,428) / 14,000

= $3.30 per equip work hour

Cost of Ownership

Repairs = (47,600 x 0.65) / 14,000 hours
= $2.21 per equip work hour

Insurance, taxes, storage = (47,600 x 0.06 per year x 7 yr) / 14,000
= 1.43 per equip work hour

Total fixed cost = Interest + Depreciation + Repairs
+ Insurance taxes & storage
= 0.95 + 3.3 + 2.21 + 1.43
= $7.89 per work hour

Tire depreciation = $4,500 / 4,000 hr = $1.13 per equip work hour

Tire repair = 0.12 X $1.13 = 0.14 per equip work hour

Cost of Ownership

Fuel cost = 150 hp x 0.06 gal per hp per hour x 0.62 (power utilization)
x 0.70 (use factor) x $1.1 per gallon
= $4.3 per equip work hour

Lubrication cost = 4 x $1.25 = $5

Lubrication labor = 2 work hours x $16.5 per work hour = $33

Lubrication cost per hour = (5 + 33) / 150 hr = 0.25 per equip work hour

Total operating cost = tire depreciation + tire repair + fuel cost
+ lubrication cost
= 1.13 + 0.14 + 4.30 + 0.25
= $5.82 per work hour

Total ownership cost = fixed cost + operating cost
= 7.89 + 5.82
= $13.71 per equipment work hour

Renting versus Buying

- Should be based upon cost
- Should be rented when needed for a short time

Mobilization

- Cost to:
  - Transportation to and from jobsite
  - Setup / erection / dismantling

Read Chapter 5
Cost of Construction Labor and Equipment

- Book
  - Peurifoy, R.L. and Oberlender, G.D. (2002). Estimating Construction Costs. McGraw-Hill.

Thank You